

Arpita Verma

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SUMMARY

Computer Science undergraduate specializing in machine learning, computer vision, and Linux-based network systems. Experienced in building end-to-end ML pipelines, from data collection to real-time deployment, and testing 4G/5G network architectures through open-source infrastructure projects.

EDUCATION

Thapar Institute of Engineering & Technology
Bachelor of Engineering in Computer Science

2023 – 2027
CGPA: 7.68

EXPERIENCE

LFX Mentorship – Magma Core (Linux Foundation)

May 2026 – Present
Remote

Open Source Mentee

- Contributed to Magma Core, an open-source 4G/5G mobile network platform, as part of the Linux Foundation's LFX Mentorship program
- Deployed and configured Magma using Docker and Linux virtual machines to simulate mobile network infrastructure
- Tested network registration and session flows using Wireshark, UERANSIM, and PacketRusher to validate 4G/5G functionality
- Collaborated with mentors via GitHub and Slack to understand and contribute to Magma's microservices architecture

PROJECTS

Cyberattack Detection & Response Agent | *Python, XGBoost, PyTorch, Gymnasium*

[GitHub](#)

- Built a two-stage system that classifies network traffic into 15 attack types (DDoS, Port Scan, SQL Injection, etc.) and automates a response action (Allow, Block, Alert, Isolate) for each flow
- Trained an XGBoost classifier on the CICIDS2017 dataset (2.27M+ traffic records across 8 days), achieving **99.64% test accuracy**
- Built to apply the full ML lifecycle end-to-end from preprocessing an imbalanced real-world security dataset to reinforcement learning and deployment as a working Gradio app.

Hand Gesture-Controlled Gameplay System | *Python, OpenCV, YOLOv8, MediaPipe, PyAutoGUI*

[GitHub](#)

- Trained and benchmarked YOLOv8n/s via transfer learning for 3-class hand gesture detection, achieving **99.4% mAP@50** and **97.7% recall** on a custom 394-image dataset
- Built a real-time CV pipeline mapping webcam gesture detection to keyboard input, using priority-based action resolution and cooldown logic
- Built a reproducible data pipeline with fixed-seed train/val splits and benchmarked model size vs. speed/accuracy tradeoffs to select the optimal deployment model

SKILLS

Languages : Python, C++, C

Engineering : Data Structures & Algorithms, OOP, DBMS

Machine Learning : Scikit-learn, XGBoost, YOLO, OpenCV, MediaPipe

Tools : Git, GitHub, Linux, Docker, VS Code, MATLAB

ACHIEVEMENTS

Cisco Networking Academy Certification – Network Defense (2026), Ethical Hacker (2026)

Open-source contributions: LFX Mentorship (Linux Foundation), GSSoC'26 contributor, OAIEDU (edge-case handling fixes for Discord Quiz Bot)